

Machine learning engineer who moves fluidly between research and production, building systems that address complex prediction problems at scale. Published researcher in environmental science bringing statistical rigor and cross-domain perspective to applied machine learning and software engineering.

Professional Experience

- Aquabyte**
Senior Machine Learning Engineer
Machine Learning Engineer

New York, NY
December 2024 – present
June 2022 – December 2024

 - Developed and deployed sea lice detection system processing ~100K images across 400+ fish farm pens daily, automating a labor-intensive monitoring task mandated by Norwegian regulations; evolved PyTorch ensemble through 3 model generations, achieving >0.7 F1 on challenging underwater small object detection task
 - Designed and automated statistical aggregation pipeline combining model detections with minimal human QA to generate daily unbiased estimates of mean lice/fish, reducing lice product COGS by ~90%; validated accuracy through field trials
 - Developed and deployed 16-day lice forecasting model with graph features and intervention simulation to 400+ pens with an interactive API, enabling customers to proactively manage lice outbreaks and optimize treatment timing
 - Architected distributed ML inference backend for fleet of vision models, scaling throughput from <100K to >2M images/day
 - Key tech:** PyTorch, YOLOv9, DinoV2, Neural Fields, Transformers, LightGBM, MLFlow, ONNX, TensorRT, Docker, Terraform
- Indigo**
Soil Data Research Intern, Carbon Experimentation

Boston, MA
June – August 2021

 - Developed generative Bayesian models to synthesize soil datasets from heterogeneous published and proprietary sources, enabling simulation of soil carbon responses to management practices across diverse geographies
 - Performed quantitative analysis of carbon sequestration potential and risk-reward tradeoffs across agricultural land use scenarios to identify high-value carbon market opportunities
 - Key tech:** NumPy, pandas, scikit-learn, Plotly

Research Experience

- New York University, Department of Environmental Studies**
Graduate Research Assistant, McDermid Lab

New York, NY
July 2020 – September 2022

 - Developed end-to-end geospatial deep learning pipeline to infer flood extent and smallholder cropland distribution from Sentinel-1 SAR and Sentinel-2 multispectral time series across sub-Saharan Africa
 - Published geospatial causal analysis localizing and quantifying the impact of flood events on food security outcomes (IPC rating) across sub-Saharan Africa from 2009–2020 using panel regression and Granger causality testing [1]
- Yale School of the Environment**
Research Assistant, The Bradford Lab

New Haven, CT
January 2016 – August 2020

 - Conducted meta-analysis (data collection, statistical analysis) of long-term experiments to quantify the mediating effect of soil organic carbon on the relationship between conservation agriculture practices and crop yield stability

Publications

[1] C. Reed *et al.*, “The impact of flooding on food security across Africa,” *Proceedings of the National Academy of Sciences*, vol. 119, no. 43, p. e2119399119, 2022, doi: [10.1073/pnas.2119399119](https://doi.org/10.1073/pnas.2119399119).

Education

- New York University**
Master of Science, Data Science

New York, NY
2022

Relevant Coursework: Probability & Statistics for Data Science • Optimization & Computational Linear Algebra • Machine Learning • Deep Learning • Inference & Representation • Forecasting Time Series Data
- Yale University**
Bachelor of Science, Environmental Studies *with distinction, cum laude*

New Haven, CT
2019

Skills

Programming	Python, SQL, Bash
ML Tools	PyTorch, scikit-learn, NumPy, pandas, ONNX, TensorRT, FiftyOne, Hydra, MLFlow, Weights & Biases
Databases	PostgreSQL, Snowflake
DevOps	Git, Docker, Terraform, dbt, Airflow, AWS
Statistics	Machine learning, deep learning, computer vision, time series, probabilistic models, Bayesian inference, hypothesis testing, A/B testing, generative models
Other	Research, writing, public speaking